Exercise Sheet 2: Investment and Asset Prices

Solutions

Brigitte Hochmuth
Department of Economics
University of Copenhagen

Problem 1 (Group 1). Asset prices and valuation

(a) No-arbitrage requires the bond price times the gross real return to equal the certain payoff:

$$(1+r)P_t^B = 1.$$

Hence

$$P_t^B = \frac{1}{1+r}.$$

(b) At $r = 0.04$,

$$PV = \frac{30}{1.04} + \frac{40}{1.04^2} + \frac{60}{1.04^3} \approx 28.85 + 36.98 + 53.34 = 119.17.$$

At $r = 0.07$,

$$PV = \frac{30}{1.07} + \frac{40}{1.07^2} + \frac{60}{1.07^3} \approx 28.04 + 34.94 + 48.98 = 111.95.$$

The payment furthest in the future is most affected because it is discounted for the largest number of periods.

(c) Required expected return on equity equals $1 + r + \varepsilon$:

$$\frac{D_t^e + V_{t+1}^e}{V_t} = 1 + r + \varepsilon.$$

Solving for today's share price gives

$$V_t = \frac{D_t^e + V_{t+1}^e}{1 + r + \varepsilon}.$$

(d) Baseline:

$$V_t = \frac{6}{0.03 + 0.04 - 0.02} = \frac{6}{0.05} = 120.$$

With $r = 0.05$,

$$V_t = \frac{6}{0.05 + 0.04 - 0.02} = \frac{6}{0.07} \approx 85.71.$$

A higher real interest rate lowers the stock price because future dividends are discounted more heavily.

(e) Rearranging the Gordon formula,

$$\frac{D_t^e}{V_t} = r + \varepsilon - g \qquad \Longrightarrow \qquad g = r + \varepsilon - \frac{D_t^e}{V_t}.$$

Hence
$$g = 0.03 + 0.04 - 0.015 = 0.055 = 5.5\%.$$
This is well above a long-run real GDP growth rate of 2.5%. The implied dividend growth looks high relative to the aggregate economy, so the valuation is difficult to justify with a simple constant-growth fundamental model.

Problem 2 (Group 2). Tobin's q and business investment with depreciation

(a) Differentiate the objective with respect to I_t:
$$-1 - c'(I_t) + q_t = 0.$$
Since
$$c'(I_t) = a(I_t - \delta K_t),$$
we get
$$q_t = 1 + a(I_t - \delta K_t).$$

(b) Solving for I_t,
$$I_t = \delta K_t + \frac{q_t - 1}{a}.$$
The first term, δK_t, is replacement investment. Replacement investment still has the direct purchase cost I_t; the adjustment cost is zero only at replacement. The second term, $(q_t - 1)/a$, is net investment driven by Tobin's q.

(c) In steady state, $K_{t+1} = K_t$, so
$$K_t = (1-\delta)K_t + I^* \implies I^* = \delta K^*.$$
Substitute this into the first-order condition:
$$q^* = 1 + a(\delta K^* - \delta K^*) = 1.$$
This is natural because in the long run an installed unit of capital should be worth its replacement cost. If q were persistently above or below 1, firms would keep expanding or reducing the capital stock.

(d) Period 0:
$$I_0 = 0.05 \cdot 100 + \frac{1.2 - 1}{0.5} = 5 + 0.4 = 5.4.$$
Then
$$K_1 = 0.95 \cdot 100 + 5.4 = 100.4.$$
Period 1:
$$I_1 = 0.05 \cdot 100.4 + \frac{1.1 - 1}{0.5} = 5.02 + 0.2 = 5.22.$$
Then
$$K_2 = 0.95 \cdot 100.4 + 5.22 = 95.38 + 5.22 = 100.60.$$

(e) Firms can change current investment spending immediately, but capital is a stock variable that changes only through the accumulation equation. A one-period investment response therefore changes the stock gradually.

Problem 3 (Group 3). Average q, marginal q, and investment subsidies

(a) Average q is

$$q^{avg} = \frac{V(K)}{K} = \frac{AK^{\alpha}}{K} = AK^{\alpha-1}.$$

Marginal q is

$$q^m = V'(K) = \alpha AK^{\alpha-1}.$$

Therefore

$$q^m = \alpha q^{avg}.$$

Since $0 < \alpha < 1$, marginal q is below average q.

(b) At $K = 100$,

$$q^{avg} = 4 \cdot 100^{-0.2} \approx 1.592,$$

$$q^m = 0.8 \cdot 1.592 \approx 1.274.$$

They differ by about

$$1.592 - 1.274 = 0.318.$$

Average q is about 25% above marginal q.

(c) Treat q_t^m as evaluated at the current capital stock and taken as given when choosing current investment. This is the local q-theory approximation. With subsidy s, the firm's dividend is

$$D_t = \Pi_t - (1-s)I_t - c(I_t).$$

Differentiating with respect to I_t gives

$$-(1-s) - c'(I_t) + q_t^m = 0.$$

Since $c'(I_t) = aI_t$,

$$q_t^m = (1-s) + aI_t.$$

Thus

$$I_t = \frac{q_t^m - (1-s)}{a}.$$

(d) Without subsidy,

$$I_t = \frac{1.274 - 1}{0.1} = 2.74.$$

With $s = 0.10$,

$$I_t = \frac{1.274 - 0.9}{0.1} = 3.74.$$

The subsidy raises investment by

$$3.74 - 2.74 = 1.00.$$

(e) Decreasing returns imply that the value of one extra unit of capital can be much lower than the average value of all existing capital. Policy can create another wedge because a subsidy lowers the marginal cost of new capital even if the stock-market valuation of existing capital does not move much.

Problem 4 (Group 4). Housing, taxation, and construction

(a) The term $r(1-\tau)$ is the after-tax financing or opportunity cost of housing, δ is depreciation and maintenance cost, and s is the property-tax cost. A lower τ raises the after-tax return on financial wealth, so an owner using accumulated wealth gives up more by holding housing. For a mortgaged owner, a lower τ also reduces the tax value of the interest deduction, so the after-tax borrowing cost rises.

(b) In short-run equilibrium, demand equals the predetermined stock:

$$H_t = H_t^d = \frac{\eta Y_t}{v p_t^H}.$$

Solving for price gives

$$p_t^H = \frac{\eta Y_t}{v H_t}.$$

Insert this into housing supply:

$$I_t^H = k\left(\frac{\eta Y_t}{vPH_t} - 1\right) + \delta H_t.$$

Using accumulation,

$$H_{t+1} = (1-\delta)H_t + I_t^H = H_t + k\left(\frac{\eta Y_t}{vPH_t} - 1\right).$$

(c) In steady state, $H_{t+1} = H_t = H^*$, so

$$0 = k\left(\frac{\eta Y}{vPH^*} - 1\right).$$

Hence

$$H^* = \frac{\eta Y}{vP}.$$

Then

$$p^{H*} = \frac{\eta Y}{vH^*} = P, \qquad I^{H*} = \delta H^*.$$

Long-run price equals construction cost because otherwise housing investment would stay above or below replacement.

(d) First compute

$$v = 0.03\left(1 - \frac{1}{3}\right) + 0.025 + 0.005 = 0.02 + 0.025 + 0.005 = 0.05.$$

Then

$$H^* = \frac{0.2 \cdot 100}{0.05 \cdot 2} = \frac{20}{0.1} = 200.$$

Also,

$$p^{H*} = P = 2, \qquad I^{H*} = 0.025 \cdot 200 = 5.$$

(e) After the property-tax increase,

$$v' = 0.03\left(1 - \frac{1}{3}\right) + 0.025 + 0.015 = 0.06.$$

New long-run housing stock:

$$H^{*\prime} = \frac{0.2 \cdot 100}{0.06 \cdot 2} = \frac{20}{0.12} = 166.67.$$

New long-run housing investment:

$$I^{H*\prime} = 0.025 \cdot 166.67 \approx 4.17.$$

Immediately after the tax change, the stock is still $H = 200$, so

$$p^H = \frac{0.2 \cdot 100}{0.06 \cdot 200} = \frac{20}{12} = 1.67.$$

Housing q is

$$q^H = \frac{1.67}{2} = 0.833.$$

Short-run housing investment is

$$I^H = 10(0.833 - 1) + 0.025 \cdot 200 = -1.67 + 5 = 3.33.$$

The higher property tax raises user cost, lowers the desired long-run housing stock, and lowers long-run replacement investment. In the short run the stock is fixed, so the adjustment comes through a lower house price, $q^H < 1$, and construction below replacement.

Problem 5 (joint problem). A discount-rate and user-cost shock: business vs housing investment

(a) *Business investment.* Before the fall in r,

$$V_0 = \frac{5}{0.04 + 0.03 - 0.02} = \frac{5}{0.05} = 100.$$

So

$$q_0 = \frac{100}{100} = 1, \qquad I_0 = 0.05 \cdot 100 + \frac{1-1}{0.5} = 5.$$

After the fall to $r = 0.03$,

$$V_1 = \frac{5}{0.03 + 0.03 - 0.02} = \frac{5}{0.04} = 125.$$

Hence

$$q_1 = \frac{125}{100} = 1.25, \qquad I_1 = 0.05 \cdot 100 + \frac{1.25 - 1}{0.5} = 5 + 0.5 = 5.5.$$

(b) *Housing investment.* Before the fall in v,

$$p_0^H = \frac{0.2 \cdot 100}{0.05 \cdot 200} = 2, \qquad q_0^H = \frac{2}{2} = 1.$$

Thus

$$I_0^H = 10(1 - 1) + 0.025 \cdot 200 = 5.$$

After the fall to $v = 0.04$,

$$p_1^H = \frac{0.2 \cdot 100}{0.04 \cdot 200} = 2.5, \qquad q_1^H = \frac{2.5}{2} = 1.25.$$

Thus

$$I_1^H = 10(1.25 - 1) + 0.025 \cdot 200 = 2.5 + 5 = 7.5.$$

(c) Business investment rises from 5.0 to 5.5, so the level change is +0.5 and the percentage change is 10%. Housing investment rises from 5.0 to 7.5, so the level change is +2.5 and the percentage change is 50%. In this calibration, housing investment responds more strongly.

(d) First, the mapping from q into investment differs across sectors because adjustment costs and supply elasticities differ. Second, housing prices can react sharply when the existing stock is fixed in the short run, while business investment is often slowed by planning, financing, and installation frictions.

Optional extension (harder). Tax change during a recession

(a) With $v = 0.06$ and $Y = 100$,

$$H^{*\prime} = \frac{0.2 \cdot 100}{0.06 \cdot 2} = 166.67, \qquad p^{H*\prime} = P = 2,$$

and

$$I^{H*\prime} = 0.025 \cdot 166.67 = 4.17.$$

(b) In period 1, $Y_1 = 95$, $v_1 = 0.06$, and $H_1 = 200$. Therefore

$$p_1^H = \frac{0.2 \cdot 95}{0.06 \cdot 200} = \frac{19}{12} = 1.583.$$

Housing q is

$$q_1^H = \frac{1.583}{2} = 0.792.$$

Investment is

$$I_1^H = 10(0.792 - 1) + 0.025 \cdot 200 = -2.083 + 5 = 2.917.$$

The next housing stock is

$$H_2 = (1 - 0.025)200 + 2.917 = 195 + 2.917 = 197.917.$$

(c) Baseline price is

$$p_0^H = \frac{0.2 \cdot 100}{0.05 \cdot 200} = 2.$$

Tax/user-cost change first, holding $Y = 100$:

$$p_{tax}^H = \frac{0.2 \cdot 100}{0.06 \cdot 200} = 1.667.$$

So the tax/user-cost component is

$$1.667 - 2 = -0.333.$$

Then the income fall, holding $v = 0.06$:

$$p^H_{joint} = \frac{0.2 \cdot 95}{0.06 \cdot 200} = 1.583.$$

So the income component is

$$1.583 - 1.667 = -0.083.$$

The total fall is

$$1.583 - 2 = -0.417.$$

Thus about

$$\frac{0.333}{0.417} \approx 80\%$$

of the immediate price fall is due to the tax/user-cost change and about 20% is due to the income shock under this decomposition.

(d) In the model, the policymaker has a point about timing: the tax increase and recession both reduce housing demand, so doing both simultaneously produces a larger short-run fall in prices and construction. However, if the tax increase is permanent, deferring it changes the transition path but not the new long-run housing stock implied by the higher user cost. A full policy evaluation would also require fiscal-revenue and distributional effects, which are outside this simple housing block.

(e) Examples include credit constraints, expectations of future capital gains, heterogeneous households, regional land-supply restrictions, construction-sector capacity constraints, and banking-sector feedbacks from falling collateral values.